Mathematical Modeling Assignment 2

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1 Part A: Population model of rabbits and foxes

A rabbit population and a fox population live in contact. The foxes feed on the rabbits, which will limit the growth of the rabbit population and facilitate the growth of the fox population. A scientist wants to model the population development.

The rabbit population R is modelled using an intrinsic growth rate of $r_R = 2$ and a maximum sustainable population K_R. This leads to a growth rate g_R:

$$g_R = r_R\, R\Big(1 - \tfrac{1}{K_R} R\Big).$$

The growth of the rabbits is further limited by rabbits being killed by foxes. This is modelled as

$$h_{FR} = r_R\, \alpha\, R\, F.$$

The α is estimated to be $\alpha = 0.01$. Here F denotes the fox population. The rabbit population is reduced when F becomes large.

In this simple model the fox population is limited by death from natural causes. The loss rate of the foxes is estimated as:

$$l_F = b_F * F$$

with a coefficient $b_F = \frac{1}{4}$.

The fox population F can only grow by feeding on rabbits. The growth is affected by the number of rabbits and the number of foxes. The larger both populations the faster the fox population will grow:

$$g_F = \tfrac{1}{B}\, R\, F.$$

It is estimated $B = 1500$.

1. Determine for which values of K_R the two species can coexist

Use the five-step method to determine the equilibria of the system and their stability. Model the problem as a dynamical system with a suitable number of state variables and a system of differential equations. Select the appropriate signs for the terms, such that they affect the population in the way described in the text.

Specify the state space. Draw a graph which has the state variables on the axes. Include the lines for which individual components of the vector field vanish. How many

equilibrium points do you count? If you change the value of K_R, does the number of equilibria change? For which values of K_R can foxes and rabbits coexist?

Include velocity vectors for a few key locations on these lines. The actual length of the vectors is not critical, only the direction is important. Based on general considerations, give a few representative vectors for the remainder of the state space, away from the lines you have drawn. Explain your reasoning for drawing the vectors in the sketch. You can draw this on paper and include as a scan or photo into your report.

Which equilibria are stable, which are unstable and for which equilibria is your investigation inconclusive?

We start by identifying all the variables listed in this problem.

Symbol	Description	Type
State variables		
R	Rabbit population	Continuous
F	Fox population	Continuous
Growth rates		
g_R	Net growth rate of rabbits	Continuous
g_F	Net growth rate of foxes	Continuous
h_{FR}	Rate of rabbits killed by foxes	Continuous
l_F	Natural loss (death) rate of foxes	Continuous
Parameters known or to be tested		
r_R	Intrinsic per-capita growth rate of rabbits	Parameter
K_R	Carrying capacity for rabbits	Parameter
α	Parameter for rabbits killed rate	Parameter
b_F	Natural death-rate coefficient of foxes	Parameter
B	Parameter for fox growth rate	Parameter

Table 1: All variables and parameters in the rabbit–fox model.

The next step now is to formulate a relation between the variables, which is in our case the differential equation system. From the definitions in the table, the two–species interactive model is then

$$\boxed{\begin{aligned} \frac{dR}{dt} &= g_R(R,F) = r_R\,R\Big(1-\tfrac{R}{K_R}\Big) \;-\; r_R\alpha\,R\,F, \\ \frac{dF}{dt} &= g_F(R,F) = \tfrac{1}{B}\,R\,F \;-\; b_F\,F. \end{aligned}}$$

Here $R(t)$ and $F(t)$ denote the rabbit and fox populations at time t. Identity that this is an dynamic autonomous system since the differential equations do not depend on time t. And our objective is to determine for which values of the capacity parameter K_R such that the two species can coexist. Next we need to find the equilibria of the system, we first plug in all the given parameters and get :

$$\frac{dR}{dt} = 2\,R\Big(1 - \frac{R}{K_R}\Big) - 0.02\,R\,F,$$

$$\frac{dF}{dt} = \frac{1}{1500}\,R\,F - \frac{1}{4}\,F\,.$$

An equilibrium should satisfy $\frac{dR}{dt} = \frac{dF}{dt} = 0$, factor F in the second equation we get :

$$\dot{F} = 0 \quad \Longrightarrow \quad F\Big(\tfrac{1}{1500}R - \tfrac{1}{4}\Big) = 0 \quad \Longrightarrow \quad F = 0 \quad \text{or} \quad R = 375.$$

Substitute $F = 0$ into $\dot{R} = 0$:

$$2R\Big(1 - \tfrac{R}{K_R}\Big) = 0 \quad \Longrightarrow \quad R = 0 \quad \text{or} \quad R = K_R.$$

The two equilibria found here are:

$$\boxed{(R, F) = (0, 0), \quad (R, F) = (K_R, 0)}$$

If $R = 375$ then sub into the first equation:

$$2\,(375)\Big(1 - \tfrac{375}{K_R}\Big) - 0.02\,(375)\,F = 0.$$

Divide by 375:

$$F = \frac{2}{0.02}\Big(1 - \tfrac{375}{K_R}\Big) = 100\Big(1 - \tfrac{375}{K_R}\Big).$$

Hence the interior equilibrium is

$$\boxed{(R, F) \;=\; \Big(375,\; 100\,(1 - \tfrac{375}{K_R})\Big).}$$

To make sure R and F strictly non-negative, we must make sure $K_R > 375$. Summarizing above results we conclude:

- For $K_R \leq 375$ there are only two nonnegative equilibria

$$(R, F) = (0, 0), \quad (R, F) = (K_R, 0).$$

- For $K_R > 375$ there are three nonnegative equilibria

$$(R, F) = (0, 0), \quad (R, F) = (K_R, 0), \quad (R, F) \;=\; \Big(375,\; 100\,(1 - \tfrac{375}{K_R})\Big).$$

For the two species to coexist, we must have the internal equilibrium asymptotically stable. Lets check this using the eigenvalue method:

We study

$$\begin{cases} \dfrac{dR}{dt} = 2\,R\Big(1 - \dfrac{R}{K_R}\Big) \;-\; 0.02\,R\,F, \\ \dfrac{dF}{dt} = \dfrac{1}{1500}\,R\,F \;-\; \dfrac{1}{4}\,F. \end{cases}$$

The Jacobian matrix is

$$J(R,F) = \begin{pmatrix} 2 - \dfrac{4\,R}{K_R} - 0.02\,F & -0.02\,R \\ \dfrac{1}{1500}\,F & \dfrac{1}{1500}\,R - \dfrac{1}{4} \end{pmatrix}.$$

1. At $(R,F) = (0,0)$

$$J(0,0) = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{4} \end{pmatrix}, \quad \lambda_1 = 2 > 0,\ \lambda_2 = -\tfrac{1}{4} < 0.$$

One positive eigenvalue, therefore *not stable*

2. At $(R,F) = (K_R, 0)$

$$J(K_R,0) = \begin{pmatrix} -2 & -0.02\,K_R \\ 0 & \frac{K_R}{1500} - \frac{1}{4} \end{pmatrix}, \quad \lambda_1 = -2,\ \lambda_2 = \frac{K_R}{1500} - \frac{1}{4}.$$

- If $K_R < 375$, then $\lambda_2 < 0$: both eigenvalues negative $\Longrightarrow$ *stable*
- If $K_R > 375$, then $\lambda_2 > 0$: one positive, one negative $\Longrightarrow$ *not stable*

3. At $(R,F) = (375,\ 100\,(1 - \frac{375}{K_R}))$

For $K_R > 375$, we checked that: $\operatorname{tr} J < 0$ and $\det J > 0$, so both eigenvalues have negative real part $\Longrightarrow$ *asymptotically stable* . This means the two species are able to coexist.

We next plot the vector field with state vectors with different K_R values: (Note that the vectors are normalized!)

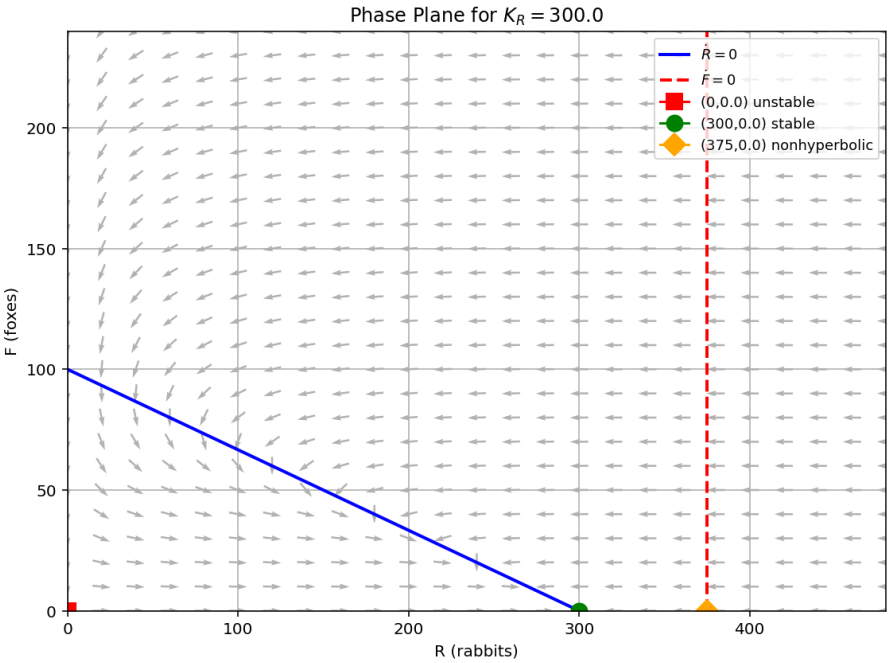

Figure 1: Vector field with $K_R = 300$.